

Les Limites

EXERCICE 1

1. $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x - 1}$

2. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{2x - 8}$

3. $\lim_{x \rightarrow -3} \frac{x^2 + 5x + 6}{2x + 6}$

4. $\lim_{x \rightarrow 0} \frac{x^3 + x^2}{4x^2 - 2x^2}$

5. $\lim_{x \rightarrow \frac{1}{3}} \frac{9x^2 + 9x - 4}{3x - 1}$

6. $\lim_{x \rightarrow 1} \frac{x^3 + x - 2}{x^2 - 1}$

7. $\lim_{x \rightarrow 2} \frac{x^3 - 2x^2 + x - 2}{x^2 - 3x + 2}$

8. $\lim_{x \rightarrow 1} \frac{2x^4 + x - 3}{x - 1}$

9. $\lim_{x \rightarrow -1} \frac{x^4 + 2x^3 + 3x^2 - 2}{x^2 - 1}$

10. $\lim_{x \rightarrow 2} \frac{x^3 + 2x^2 - 16}{x - 2}$

EXERCICE 2

1. $\lim_{x \rightarrow 2} \frac{\sqrt{3x+3} - 3}{x^2 - 4}$

2. $\lim_{x \rightarrow 0} \frac{\sqrt{1-x} - 1}{x}$

3. $\lim_{x \rightarrow 4} \frac{\sqrt{x-1} - \sqrt{3}}{x^2 - 16}$

4. $\lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3}{\sqrt{x+2} - 2}$

5. $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} - \sqrt{2x}}{x - 2}$

6. $\lim_{x \rightarrow -2} \frac{x + \sqrt{x+6}}{x + 2}$

7. $\lim_{x \rightarrow 0} \frac{\sqrt{x+3} - \sqrt{4x+3}}{x}$

8. $\lim_{x \rightarrow 2} \frac{\sqrt{2x} - 2}{\sqrt{x+1} - \sqrt{2x-1}}$

9. $\lim_{x \rightarrow 1} \frac{\sqrt{x+3} + \sqrt{x+8} - 5}{x - 1}$

10. $\lim_{x \rightarrow 2} \frac{\sqrt{x+2} + \sqrt{x+5} - 5}{\sqrt{x-1} + \sqrt{x+2} - 3}$

EXERCICE 3

1. $\lim_{x \rightarrow 0} \frac{3x - 1}{x^2}$

2. $\lim_{x \rightarrow 1^+} \frac{1 + x^2}{(x-1)(4-x)}$

3. $\lim_{x \rightarrow 3^+} \frac{6 + x^2}{6 - 2x}$

4. $\lim_{x \rightarrow 2^+} \frac{x + x^2}{x^2 - 4}$

5. $\lim_{x \rightarrow 1^+} \frac{x + 2}{x^2 - 3x + 2}$

6. $\lim_{x \rightarrow 2^+} \frac{\sqrt{x-2}}{x-2}$

7. $\lim_{x \rightarrow 1} \frac{2x}{|x-1|}$

8. $\lim_{x \rightarrow 2} \frac{x^2 + |x| - 4}{x^2|x| - |x|}$

9. $\lim_{x \rightarrow 1^+} \frac{x+1}{\sqrt{x-1}}$

10. $\lim_{x \rightarrow 0^+} \frac{x^2 - 2}{\sqrt{x+4} - 2}$